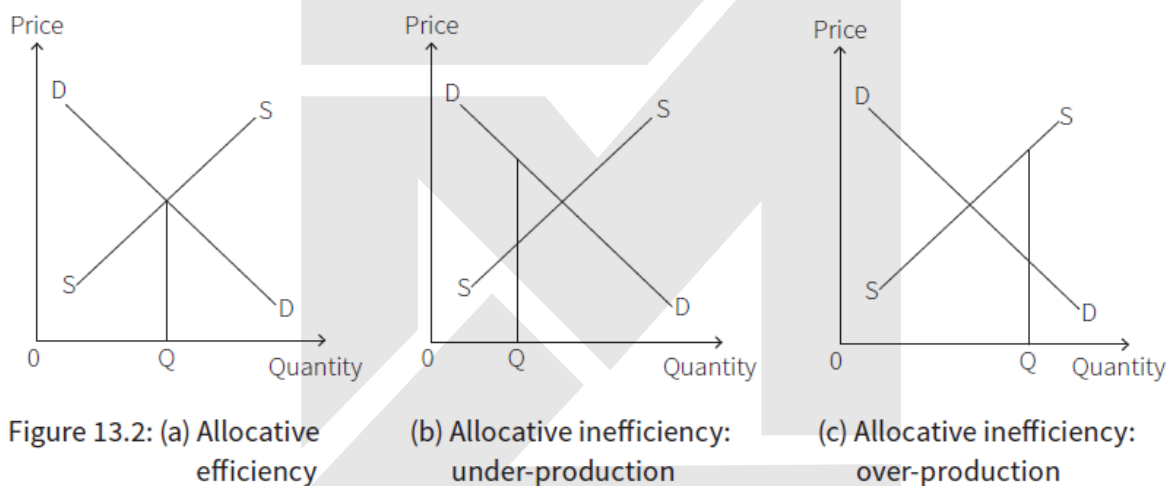


Market failure

Efficiency

Allocative efficiency occurs when resources are allocated in a way that maximizes consumers' satisfaction. This means that firms produce the products that consumers demand, in the right quantities.

Figure 13.2 a shows allocative efficiency being achieved with supply matching consumers' demand. In contrast, Figures 13.2 b and 13.2 c depict allocative inefficiency. In the case of 13.2 b, there are too few resources being devoted to the product, which results in a shortage. In 13.2 c, too many resources are allocated for producing the product and there is a surplus.



Market forces, by changing prices, should eliminate shortages and surpluses, and move markets towards allocative efficiency. Competition can play a key role in this process. This is because in a competitive market, a firm has an incentive to be allocatively efficient in the form of profit. It also has a threat of punishment in the form of the risk of going out of business if it is not allocatively efficient. If it is more responsive to the needs of consumers as compared to its rivals, it should gain a larger market share and earn high profits at least for a while. In contrast, if it does not produce commodities demanded by consumers, it will lose sales to rivals and may be driven out of the market.

Productive efficiency: A firm is said to be productively efficient when it produces at the lowest possible cost per unit. Again, in a competitive market, a firm has both an incentive and a threat of punishment which should drive it towards being

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Market failure

productively efficient. If it can drive its costs down to the lowest possible level, it may capture more sales and gain more profit. If, however, its costs per unit are higher than its rivals, it will lose market share and possibly all of its sales.

If a firm is productively efficient, it means that it is not wasting resources. If all producers in a country are productively efficient, the economy will be able to make full use of its resources and hence will be producing on its production possibility curve.

In Figure 13.3, production point A is productively efficient. With its given resources and technology, the economy is making as many products as possible. Point B is productively inefficient as some resources are either not being used or not being put to good use. For example, some workers may be unemployed, some workers may be lying idle, and some factory and office space may be empty. Also, there may be some workers involved in jobs to which they are not best suited and the capabilities of some capital goods may not be fully exploited.

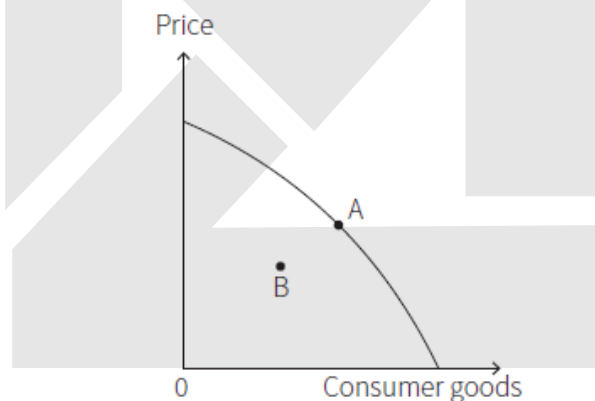


Fig. 13.3: Productive efficiency (A) and inefficiency (B)

Dynamic efficiency: Dynamic efficiency arises when resources are used efficiently, over a period of time. The profit incentive, and threat of going out of business, can encourage firms in a market system to spend money on research and development, and to innovate. Those firms that introduce new methods of production and bring out new, improved products, increase their chance of gaining high profits. Those that do not seek to keep up with new ideas to produce products and do not develop new products, run the risk of being driven out of the market.

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Market failure

Market failure occurs when freely functioning markets fail to deliver an efficient allocation of resources. The result is a loss of economic and social welfare. Market failure exists when the competitive outcome of markets is not efficient from the point of view of society as a whole. This is usually because the benefits that the free market confers on individuals or businesses carrying out a particular activity diverge from the benefits to society as a whole.

Private cost are the costs which are incurred by the producers when they produce goods and services e.g. cost of labor, raw materials, and machines and so on. Consumers also incur private costs when they buy different goods and services. The greater the production or consumption of goods and services, the higher will be the private costs to producers and consumers. These are also called internal costs in the sense that they are borne by those who produce or consume goods and services.

Private benefit are the benefits which are received by the producers when they sell goods and services which they produce. The revenue which the producer receives is the private benefit. Consumers also receive the benefits when they consume the goods and services. The utility or satisfaction which they receive is the private benefit. These benefits are also called internal benefits in the sense that they are received only by those who produce or consume goods and services.

Externalities are third party effects arising from the production and consumption of goods and services for which no appropriate compensation is paid. Externalities cause market failure if the price mechanism does not take account of the external cost and benefits of production and consumption. Externalities occur outside of the market i.e. they affect economic agents not directly involved in the production or consumption of a particular goods or service. Since externalities do not go through the market, they do not have prices attached to them.

NEGATIVE EXTERNALITIES occur when production or consumption impose external costs on third parties outside of the market for which no appropriate compensation is paid. For example consider the situations in which a business dumps the waste products from its production process in to a nearby river or in which an individual litters a park or beach. Obviously, a cost is involved in these

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actions. When the firms pollute the water, people downstream suffer consequences. They may not want to swim in or drink polluted water. They may also be unable to catch fish as before because of pollution. In the case of littering, the people who come along after our litter has cluttered the park or the beach are the ones who bear the costs. The costs of these actions is borne by people other than those who commit these actions, the creator of the costs is not the sole bearer. The costs are not internalized by the individual or firm; they are external.

Some examples are:

1. Smokers ignore the unintended but harmful impact of passive smoking on non-smokers or tribute to the growing costs of treating smoking-related diseases.
2. Acid rain from power stations
3. Air pollution from the road use and traffic congestion the external costs of drugs and alcohol abuse
4. The environmental damage cause by the intensive use of fertilizers in agriculture

POSITIVE EXTERNALITIES exist when third parties benefit from the spillover effects of production and consumption. Examples are:

1. Social returns from increase investment in education and training. If one person gets educated, it benefits the entire society. The person can work as a trained worker, or he/she can be a trained professional worker at any job.
2. Positive benefits from improvements in the quality of health care and medical research.
3. Improved health outcomes arising from vaccination program. If one gets vaccinated, it prevents others form contracting the disease from that person, thus a third party is not involved in the transactions.
4. Provision of flood protection system and installation of more effective fire safety equipment
5. The external benefits to be derived from restored historic buildings and monuments.
6. External benefits from people usage of public libraries and museums.

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SOCIAL COSTS is the total cost which a society bears as a result of production and consumption activity in society. It is the sum of private costs (costs to the producers and consumers of goods and services) and external costs (costs which the third parties bear as a result of production and consumption of others). Social cost = private cost + external cost.

SOCIAL BENEFITS are the total benefits which a society receives as a result of production and consumption activities in the society. It is the sum of private benefits (benefits received by the producers and consumers) and external benefits (benefits received by the third parties). Social benefit = private benefit + external benefit

CAUSES OF MARKET FAILURE AND GOVERNMENT INTERVENTION:

1. **Public goods:** Goods that are neither excludable nor rivalrous are called public goods or sometimes called collective consumption goods. The classic case is national defense. An army of a given size protects all the nation's citizens equally no matter how many citizens are there and whether or not a particular individual pays taxes to support it. Other examples include lighthouses, street lights, etc.

Non-rival consumption means that consumption by one person does not affect the consumption of others. Non-excludable consumption means that no one can be excluded from the consumption of public goods. As a result, no one will be willing to pay for them. So the private sector does not produce these goods which lead to market failure. In order to avoid market failure, the government provides these goods which are financed out of taxation as in public goods there will always be existence of free rider problem.

2. **Demerit goods:** Goods where social cost exceeds private costs that is existence of external costs. These are the goods which are harmful for the society and have immense external costs. Examples include cigarettes, alcohol and drugs. These are over provided and undercharged by the market economy because people are unaware of the true cost of consuming them,

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this represents information failure or lack of information (in true sense ignorance)

Cigarettes cause unpleasant smoke which is dangerous to people in the area who are forced to become passive smokers. They inflate the national health bill because both the smokers and the passive smokers get sick. But smokers will not stop, perhaps are unable to do so because they are addicted.

Too many demerit goods are demanded, so the government steps in and taxes highly in order to reduce the consumption and to raise the revenue which is needed anyway to spend treating to sick ones. The government advertises heavily to try to persuade people stop smoking and the young not to start and is seriously considering banning smoking in all public work places

3. **Merit goods:** Goods where social benefits exceeds private benefits that is existence of external benefits which are not realized due to information failure. There is a class of goods whose consumption benefits are higher for the society as compared to individuals, these goods are called merit goods e.g. education and health. When a person gets educated, the acquired knowledge and skills increase the income of individual. To society actual benefits are higher productivity, new and better products, civilized people etc. in such a case a market economy under allocates scarce resources to the provision of merit goods i.e. there is under consumption and under production in the market which results in market failure. Government can intervene to correct the situations by providing subsidies to private sector or provide themselves for e.g. state education.

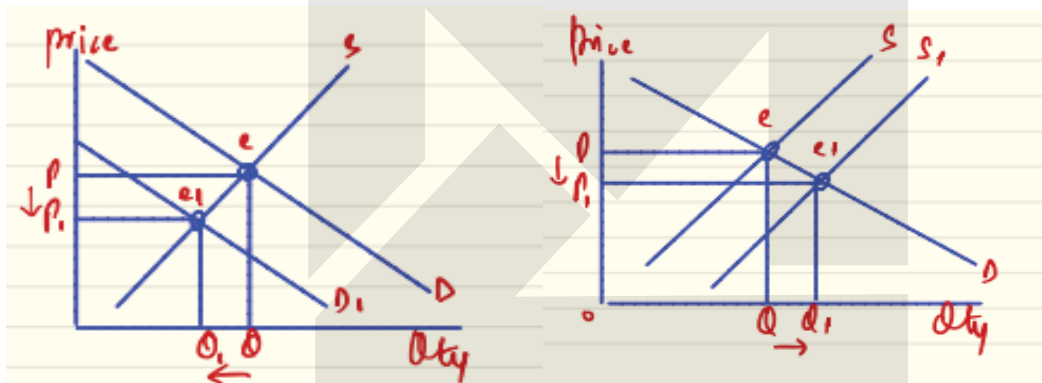
Information failure: For consumers to buy the products that will give them the highest possible satisfaction at the lowest possible prices, they have to be fully informed about the nature of the products on offer, the benefits they can receive from them and their prices. Workers need to know what jobs are on offer, the location of the workplace, the qualifications required and the remuneration they would receive. They should also be aware about the nature of jobs for which their skills are best suited.

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Similarly, producers need to know what products are in demand, where good quality raw materials can be purchased at lowest possible prices and what are the most cost-effective methods of production. If they lack this information, they will make decisions that are not in their best interests. Besides consumers paying more than required and buying products of lower quality than available, workers may end up in the wrong jobs, and producers' costs may be higher and revenues lower than possible due to information failure.

Information failure can occur in a number of ways. There may be a lack of information or inaccurate information. There may also be asymmetric information which occurs when consumers and suppliers do not have equal access to information. For instance, if a car mechanic tells a motorist that her or his car needs an expensive repair, the motorist may lack the technical knowledge to question the advice.



4. Market failure can arise due to producers having more **market power** than consumers. If one firm dominates a market, it may not be allocatively, productively or dynamically efficient. It will lack competitive pressure to respond to consumer demands, to keep its costs low and to improve its product. If it is the only firm selling the product, that is a monopoly: consumers will have no choice but to buy from it, even if the price of the product is high, the product does not meet the needs of the consumers and its quality is poor.

Abuse of market failure can also occur, when there is more than one firm producing the product. If there are, for example, five major producers in a market

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there is a risk that they may collude to reduce competition and, in effect, act as one seller. For example, they may all agree to charge the same high price. This is referred to as price fixing.

There are various ways through which governments try to promote competition and prevent firms from abusing their market power. These include removing restrictions on the entry of new firms into a market and making uncompetitive practices such as price fixing illegal. They may also stop some firms from merging, that is joining together to form one new firm, if it is thought that the merged firm will act against the interests of consumers by charging high prices and producing poor quality products.

5. **Immobility of resources:** To achieve allocative efficiency, it is necessary for resources to move from producing products that are decreasing in demand towards those which are experiencing an increase in demand. This requires resources to be both occupationally and geographically mobile. In practice, some resources may be immobile. If, for example, demand for a country's financial services might be increasing, whilst demand for its steel may be decreasing, there may be a shortage of financial services, unemployment of workers and underutilization of capital equipment if resources cannot easily move between the two.

The main measures a government can take to promote occupational mobility of labour are to improve education and to provide training in the new skills needed. Also, governments can provide investment grants to make it easier for firms to change the use of land and buildings. Geographical mobility of workers can be encouraged by making it easier for them to buy or rent housing in areas where demand for labour is high. This might be achieved by construction of more houses in such areas or by the government providing financial help for those workers who move to these locations.

Policies

Competition policy seeks to promote competitive pressures and prevent firms from abusing their market power. There are a number of ways a government might be able to do this, including prevention of mergers that it thinks will not be in the

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interest of consumers, removal of barriers to entry and exit into markets, regulation of monopolies and prohibition of uncompetitive practices. Uncompetitive practices may include, for example, predatory pricing and limit pricing. Predatory pricing involves a firm charging a price below the cost to drive a rival firm (or firms) out of the market. Limit pricing is setting the price low enough to discourage the entry of new firms into the market.

Environmental policies: Firms can be affected by a range of policies designed to improve environmental conditions. A government may place restrictions on the amount of pollutants emitted by firms into the air, sea and rivers. It may then fine any firms which exceed these limits.

Another policy, which has become more popular in recent years, is tradable permits. This involves a government issuing permits to firms, allowing them to pollute up to a certain limit and to sell part of their allocated limit, if they pollute less. The idea is that the cleanest firms will be able to sell most of their permits, whilst those who pollute the most will have to buy more of other firms' permits. This will reduce the costs of the cleanest firms, whilst raising the costs of the worst polluting firms. As a result, the cleanest firms should capture a higher market share and consequently, pollution should fall.

Regulation includes rules and laws which place restrictions on the activities of firms. Besides setting price controls, outlawing uncompetitive behavior and limiting the amount of pollution emitted by a firm, a government may regulate the target audience for the product, the quality of products and mode of staff management by firms. For example a government may pass a law banning the sale of cigarettes to children. It may also require firms to ensure that the products produced by them meet certain standards and that they allow their workers a specified number of regular holidays. In addition, it may place restrictions on timing for opening/closing of shops and control the routes that buses must follow.

As a measure to correct market failure, regulations have the advantages of being backed up by law and easily understood. The government does, however, have to check that the rules and laws are being followed and this may be difficult and expensive. Also, a regulation works only if most people agree with it. For example,

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it would be difficult to enforce a law that everyone wears a safety helmet when riding a motorcycle if such a move is opposed by most of the riders. This is because too much time and money would need to be spent on prosecuting the offenders and the government may become very unpopular.

There are a number of other problems with imposing regulations, for example they do not directly compensate those who suffer as a result of market failure and regulations may be too restrictive - reducing market flexibility and creating barriers to entry.

Nationalization and privatization: To benefit the public and to improve economic performance, a government may set up an industry or nationalize a private sector industry. Industries owned by the government are known as state-owned enterprises, public corporations, and nationalized industries. The chairman and board of managers are appointed by the government. They are responsible for the day-to-day management, but are accountable to the government. There are no shareholders in state-owned enterprises. The funds come from the government, from government approved loans and from the private sector. State-owned enterprises do not always seek to make a profit. Their prime aim is to work in the public interest.

There are a number of **advantages** that can be claimed for state-owned enterprises. Some of them are as follows:

- State-owned enterprises base their decisions on the full costs and benefits involved.
- They can be used to influence economic activity. To boost the country's output, public corporations can be directly encouraged to increase their output.
- In cases where it is practical to have only one firm in the industry, such as rail infrastructure, a state-owned enterprise would not abuse its market power.
- Ownership of a whole industry by the government makes planning and coordination easier. For example, if the state runs the train system, it can ensure that train timetables are coordinated.

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- It is important to ensure that basic industries, such as electricity and transport, survive, charge low prices and produce good quality, as other domestic industries depend on them.

There are, however, a number of **disadvantages** associated with state-owned enterprises.

- They can be difficult to manage and control. The large size of the organizations may mean that time has to be spent on meetings and communicating with staff, slowing down decision making.
- They may become inefficient, produce low quality products and charge relatively high prices, due to a lack of competition and the knowledge that they cannot go bankrupt.
- They will need to be subsidized if they are loss making. The use of tax revenue to support them has an opportunity cost - it could be used to spend on, say, training more teachers and nurses.

Concern about the performance of state-owned enterprises and increased confidence in market forces has led a number of countries to sell their state-owned enterprises, or part of their state-owned enterprises, to the private sector. Those supporting this move argue that private sector firms are likely to produce the products desired by consumers, at a low cost and offer them at low prices. This is because market forces provide an incentive for firms to be efficient in the form of profit and a threat of bankruptcy if they are inefficient. Besides low prices and high quality, privatization may result in greater choice. Freedom from government regulation may reduce administration costs and enable managers to respond more quickly to changing conditions. There may, also, be less risk of under-investment in the private sector. The funds available to a public sector firm for investment will depend on the profits it earns and its ability to convince shareholders and lenders of its success. Public corporations may be kept short of funds for investment, however successful they are, if the government wants to spend the money elsewhere.

Privatization, however, is itself criticized. There is no guarantee that private sector firms will face the full pressure of market forces. Some private sector firms may not face competition - they may be monopolies (i.e. the only firm selling

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the product). In this case, they can be inefficient, charge high prices and produce low quality products without compromising on profits. They may not take into account the total costs and benefits to the society due to their actions. For example, they may cause pollution. Privatization also reduces a government's control of the economy.

Direct provision: Most governments produce at least some goods and services that they think are essential. In some countries, governments provide affordable housing to rent. Housing, education and healthcare are seen as essential services, and some governments produce them and provide them to people free of cost or at subsidized prices. Besides being essential products, education and healthcare are also merit goods.

A merit good is one whose benefit to consumers and others is undervalued by them. As a result, they would under-consume it and so private sector firms would under-produce it. This is why some governments produce educational and healthcare services and other merit goods such as library services. To stimulate the consumption of merit goods, governments also pay private sector firms to produce them, provide information about their benefits and, in some cases, make their consumption compulsory.

There are some products that private sector firms have no incentive to produce. This is not because people do not want them, but because they know that if they are provided, they can consume them without paying. For example, it is not possible to exclude someone from enjoying the benefits of street lighting even if they are not prepared to pay for them directly. This is why governments produce them, or pay private sector firms to make them, and raise finance through taxation. It is also interesting to note that public goods are non-rival. This means that a person enjoying the product does not reduce someone else's enjoyment. An additional family moving into a town protected by sea defenses, does not reduce the defense experienced by other families.

Unfairness: Besides intervening in a market economy to correct market failure, governments also intervene on grounds of equity - that is, fairness. Income distribution can become very uneven if it is solely determined by market forces.

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Market failure

Some people will be very rich, but some will be very poor. Private sector firms will only produce those products that people are willing to buy and able to pay for. This may mean that they will not produce products needed by the poor.

A government is likely to try and ensure that everyone in the country has access to basic necessities including housing, education and healthcare. To achieve this, it can give financial assistance to the poor and provide some essential products free to consumers. In fact, in a number of countries, state education and healthcare is provided free, not only because they are merit goods, but also to make them accessible to the poor. Such Free State services are financed by taxation. Taxation and benefits may also be used to reduce income and wealth inequality.

A big difference in the income and wealth of the rich and poor, besides being unfair, can be socially divisive and can result in some workers being less productive. Some people in the country, including the elderly and the sick, may be unable to earn incomes. There may be social unrest if there is considerable income inequality. Also, if people are poor they may be less healthy, less well educated and consequently less productive.

Effectiveness of government intervention

As suggested above, government intervention can reduce market failure. There is a risk, however, that government failure may occur. It may overestimate the extent of the private benefits offered to the people by consuming merit goods and it may find it difficult to calculate the most efficient quantity of public goods to supply.

Governments can take time to make decisions and those decisions may be influenced by political factors and, in some cases, corruption. For example, a government may decide not to raise the tax on petrol, despite concerns about the environment, because it may be politically unpopular and may lose its votes.

Government intervention may also reduce economic efficiency by reducing incentives. If taxes on earned income and unemployment benefits are high, some people may be discouraged from working. High taxes on firms' profits can reduce entrepreneurs' willingness and ability to invest.

A development of effectiveness of government intervention

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Market failure

There is some debate as to whether public or private sector expenditure leads to a more efficient allocation of resources. Do households and firms make better decisions than the government? In practice, there are advantages and disadvantages of both private and public sector expenditure. A new airport, for example, could be built by the private or public sector. There may be a number of advantages in it being built by a private sector firm. The profit incentive and force of competition may imply that it will build a high quality airport at low cost and in less time.

There is a risk, however, that a private sector firm may be a monopoly and hence may not be forced to keep its costs down. Thus, it may charge a high price for building the airport. Also, a private sector firm will take only private costs and benefits into account.

Using public expenditure to build an airport may also have its own drawbacks. Knowing that the state is paying, a state-owned enterprise, or private sector firm hired by the government, may not keep its costs down. A state-owned enterprise may lack the commercial expertise to complete the project on time. There may also be delays in decision making by the government to go ahead with the project. A major benefit, however, of a major investment project being undertaken by government is that it will base its decision (as to whether to proceed with it) on the consideration of all factors involved, that is social impact, costs and benefits. It is likely to carry out a cost benefit analysis (CBA) in the first place. This involves measuring all the private costs and benefits involved. In the case of an airport, the private costs will involve the cost of the land, the cost of the labour employed to build and run the airport, and the cost of the building material and maintenance. The major private benefit is the revenue that will be earned.

After measuring the private costs and benefits, the economists carrying out a CBA, then seek to place a value on external costs and benefits. This is not an easy process. The external costs may include operation of the airport, damage to the environment, noise due to risk of accidents and traffic congestion near the airport. The external benefits may include employment in the area due to tourism and making it a more attractive as a site for domestic firms and multinational companies (MNCs).

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Market failure

When all the calculations have been made, the social costs and benefits are compared. If social costs exceed social benefits, the government will not proceed with the project. If social benefits exceed social costs, it will go ahead (if the net social benefit is greater than that on rival projects). There will still be a debate, however, on whether it is the best use of government money. Government expenditure on one item always involves a significant opportunity cost. The money could have been spent on, for example, education.



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